SHANG WU

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RESEARCH INTERESTS

I am broadly interested in human–AI interaction, AI evaluation and benchmarking, and uncertainty quantification. I develop statistical and cognitively inspired methods to evaluate and improve AI systems. My work spans Bayesian and sequential evaluation of black-box LLMs for reliable, cost-efficient benchmarking; uncertainty calibration and how LLMs express certainty in natural language; and optimal task allocation across human and AI agents under real-world constraints. I design user studies on how AI assistance shapes human reasoning, learning and reliance on deep learning systems.

EDUCATION

Ph.D. in Computer Science **09/2023 – Expected 06/2028**
University of California, Irvine
Advisor: Padhraic Smyth

M.S. in Computer Science **09/2023 - 03/2025**
University of California, Irvine | GPA 3.96/4.00

B.S. in Computer Science and Psychology with Honors **09/2020 - 03/2023**
University of California, Davis | GPA 4.00/4.00

PUBLICATIONS

Belém, C., **Wu, S.**, Yao, H., Steyvers, M., Singh, S., Smyth, P., From 'May' to 'Is': Certainty Distortion in Language Model Rewriting. *EMNLP 2026*

Wu, S., Belém, C., Fu, S., Steyvers, M., Smyth, P., How AI Assistance Affects Human Skill Development: A Study of Learning with Logic Puzzles. *HCOMP 2026*

Wu, S., Yao, H., Belém, C., Fu, S., Steyvers, M., Smyth, P., The Impact of AI Usage and Informativeness on Skill Development in Logical Reasoning. *HHAI 2026*

Longjohn, R., **Wu, S.**, Kher, S., Belém, C., Smyth, P., Bayesian evaluation of blackbox LLM behavior. *NeurIPS 2025 Workshop on Evaluating the Evolving LLM Lifecycle: Benchmarks, Emergent Abilities, and Scaling*

WORKING PAPER

Wu, S., Kher, S., Smyth, P., Learning to assign prediction tasks to agents with capacity constraints. *Under Review*

Kher, S.*, **Wu, S.***, Longjohn, R., Belém, C., Smyth, P., Sequential Bayesian Evaluation of Large Language Model Behavior *Under Review*

*Equal contribution

Wu, S., Yao, H., AI evaluation may bias perceptions: The importance of context in interpreting academic writing. *Under Review*

FELLOWSHIPS & AWARDS

HPI Research Center Fellowship	01/2025
UCI ICS Dean's Fellowship	06/2024
Thomas E. Bruzzone Scholarship	12/2022
Charles R. Bishop Award	09/2022
2021 Michael's Scholarship	01/2022

RESEARCH EXPERIENCE

Graduate Student Researcher **09/2023 - Present**
University of California, Irvine, Computer Science Department

Research Assistant
University of California, Davis, Psychology Department 06/2022 - 03/2023
University of California, Davis, Animal Science Department 02/2022 - 05/2022

SKILLS

Machine Learning · Deep Learning · Large Language Models (LLMs) · Model Evaluation & Benchmarking · Uncertainty Quantification
Statistical & Bayesian Modeling · Experimental Design · Human–AI Interaction · User Studies · Crowdsourced Data Analysis
Python (PyTorch, scikit-learn, NumPy, pandas, matplotlib, Stan) · C++ · Java · Git
English · Mandarin

TALKS & PRESENTATIONS

Wu. S, Understanding human-AI dynamics in interactive decision-making tasks, *Collective Intelligence 2025 Demos and Posters.*

TEACHING

Student Assistant - UC Davis Computer Science Department **04/2022 - 06/2022**
Course: Software Development & Object-Oriented Programming in C++.

Peer Tutor – UC Davis Academic Assistance and Tutoring Centers **10/2021 - 03/2023**
Programming, mathematics, and statistics.